

Project Prompt Log

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Project Name: *MedStorePro Inventory Management System*

Overview

This document serves as a comprehensive log of the prompts and requests made during the development of the project. It details the specific topics addressed and the problems resolved for each interaction.

Antigravity Prompts

ID	Date	Topic	Primary Prompt / Objective	Problem Solved / Outcome
1	2025-12-23	Project Health Check	<i>User requested to verify if the entire project (backend + frontend) is working as expected.</i>	System Verification: Verified the state of the application. Addressed issues with Docker Compose failing to pull images, ensuring that both the backend and frontend services could start and interact correctly.
2	2025-12-23	Debugging AWS Login 404	<i>User reported a 404 error when attempting to log in on the AWS deployed version.</i>	API Routing & Deployment: Investigated frontend API_BASE_URL configuration and backend routing in app.js. Checked AWS serverless/deployment configurations to ensure the /api/auth/login route was correctly exposed and mapped in the API Gateway.
3	2025-12-15	Bug Fix (Delete Order)	<i>User requested to fix the "Delete Order" button functionality</i>	Functional Debugging: Debugged the DELETE request failure to the local backend. Fixed the event handler for the delete action and resolved the issue with the theme toggle state persistence.

			<i>and the dark/light theme toggle.</i>	
4	2025-12-11	Docker Configuration	<i>User needed to successfully build and run the backend using Docker Compose, facing multiple errors.</i>	Environment Setup: Resolved PrismaClientInitializationError (version mismatch) and TypeScript compilation errors. Fixed port conflicts (5432/5000) by remapping ports and changing backend to port 5001.
5	2025-12-11	Docker Build Errors	<i>User encountered compilation errors during the Docker build process for the backend.</i>	Build Troubleshooting: Fixed TypeScript and Zod validation errors that were blocking the build. Corrected Prisma configuration within the Docker environment to ensure successful containerization.

Detailed Breakdown

1. Project Health Check

- **Context:** The user returned to the project and needed to ensure the development environment was still operational.
- **Key Actions:** Ran docker-compose up, checked container status, verified frontend build.

2. Debugging AWS Login 404

- **Context:** Deployment issue where the authentication endpoint was unreachable.
- **Key Actions:** Audited environment variables in pharmacy_frontend-main/src/config/environment.js and backend route definitions in routes/auth.js.

3. Fixing Delete Order & Theme Toggle

- **Context:** Frontend interactivity issues.
- **Key Actions:** Traced the HTTP request from the button click, checked network logs, and corrected the backend handler or frontend service call. Fixed CSS/JS logic for theme switching.

4. Debugging Docker Build and Ports

- **Context:** Infrastructure setup challenges.
- **Key Actions:** Analyzed Docker logs, identified port 5000 was in use, switched to 5001. Resolved critical Prisma generation steps in the Dockerfile.

5. Debugging Docker Build Errors

- **Context:** Initial containerization efforts.
- **Key Actions:** Systematically fixed compile-time errors in the backend codebase to allow for a clean Docker image build.

Cursor app prompts

#	Prompt (User Request)	Topic Area	Problem / Goal it Solved	Outcome / Deliverable
1	“implement the restock functionality on dashboard so that if we restock or change the stock quantity it also updates the quantity on database”	Frontend + Backend Integration	Stock updates were only UI-level; needed DB update for real inventory accuracy	Added restock flow + API call to update product quantity in DB
2	“in the orders the delete button is not working”	Frontend Bug Fix	Delete action not triggering correct API / failing silently	Fixed frontend delete handler to use centralized API wrapper
3	“it is now showing internal server error”	Backend Debugging	Delete endpoint failing due to schema/field mismatches	Debugged controller, fixed mismatched fields + improved error handling
4	“the delete order is still not working”	Backend Reliability	Delete still failing due to edge cases (missing items, populated vs unpopulated IDs)	Hardened delete logic, null checks, safer stock restore
5	“give the admin user access to delete order”	Authorization / RBAC	Delete route restricted; admin needed access	Updated backend route middleware to admin

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6	“try again”	Iterative De bugging	Previous fix not fully solving issue	Continued debugging + patching until delete stabiliz ed
7	“make the light theme and dark theme in working condition”	UI/UX	Theme toggle existed but not applied consistentl y	Fixed theme context and rep laced hardcoded colors with theme values
8	“the order button in the order page is not working it s giving me internal server error”	Backend + Frontend B ug Fix	Order creation failing; needed controller validation + correct client calls	Fixed createOrder handling + frontend order create call + logging
9	“for backend url to use in frontend use this https://8d40fk8n44.execute-api.us-east-1.amazonaws.com/dev ”	Deploymen t Config	Frontend not point ing to correct backend base URL	Updated environment config guidance for production API URL
10	“error while logging in”	Auth / Deploymen t Debuggin g	Deployed login fai ling; suspected URL/C ORS mismatch	Investigated API base URL and endpoint paths
11	“its still not working”	Auth / Debugging	Login still failing after prior changes	Identified missing /api segment in production base URL
12	“push the new changes to github”	Version Co ntrol	Needed to publish latest fixes	Committed and pushed chan ges to GitHub

#	Prompt (User Request)	Topic Area	Problem / Goal it Solved	Outcome / Deliverable
13	“i want to use lamda function of aws so i can use it to export data from mongodb to lambda function to S3 bucket”	AWS Architecture	Needed analytics export pipeline	Proposed Lambda → S3 export architecture and code approach
14	“i can't use any paid services”	Cost Constraints	Needed free-tier compatible approach	Recommended free-tier strategy; explained tradeoffs (NAT cost warning)
15	“i can use NAT gateway then what should i do and i can use aws free tier”	AWS Networking	Needed stable outbound IP for Atlas allowlisting	Designed VPC private Lambda + NAT + allowlist workflow
16	“read the entire project”	Codebase Understanding	Needed accurate documentation aligned with actual project code	Reviewed project structure: frontend, backend, routes/models/controllers
17	“now i want you to write a project proposal for me like this sample proposal demo”	Documentation	Needed proposal aligned to project + AWS	Generated proposal (executive summary, problem, objectives, architecture)
18	“help me implement the lambda function”	Implementation Guidance	Needed step-by-step Lambda build/deploy instructions	Provided Lambda code + env vars + packaging steps
19	“can you check is my mongodb atlas or simple mongodb”	Database Identification	Needed to know connectivity method	Explained mongodb+srv:// = Atlas, mongodb:// = standard

#	Prompt (User Request)	Topic Area	Problem / Goal it Solved	Outcome / Deliverable
20	“give me step by step guide... create s3 bucket... lambda... NAT gateway...”	AWS Implementation	Needed complete deployment steps	Provided click-by-click AWS console guide
21	“in this part... create policy... do i need to create the policy in the S3 bucket or somewhere else?”	IAM	Confusion between IAM policy vs bucket policy	Clarified: IAM policies in IAM; bucket policies in S3
22	“i got this error while creating the role”	IAM Restrictions	Lab account blocked iam:CreateRole	Recommended using existing roles + bucket policy workaround
23	“there is no permission tab in the create function”	Console Navigation	UI confusion in Lambda console	Guided to “Change default execution role” & config permissions
24	“in runtime there is no node.js 18.x...”	Runtime Selection	Needed correct Lambda runtime	Chose Node.js 20.x as stable option
25	“should i not select anything in the additional configurations part”	Lambda Setup	Avoid unnecessary settings	Recommended leaving most off; enable VPC only
26	“these are the stuff in the additional configuration what should i do”	Lambda Setup	Unsure which checkboxes to use	Guided: enable VPC; keep Function URL/KMS/code signing off
27	“i have created the function now what should i do”	End-to-End Setup	Needed next steps after Lambda creation	Ordered steps: S3, permissions, NAT, allowlist, env vars, test
28	“what should i put in place of the placeholders in step 3?”	IAM/Bucket Policy	Needed correct ARNs and bucket name	Explained how to extract account ID + role ARN

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29	“is my lambda attached to the vpc”	Verification	Confirm VPC association	Verified Lambda attached to VPC + private subnets
30	“i have not created a NAT gateway yet”	Networking	Lambda couldn't reach Atlas without NAT	Guided NAT creation + route table config
31	“what should i select in the create nat gateway”	NAT Setup	Needed correct NAT settings	Recommended zonal + public + manual EIP + public subnet
32	“which public subnet should i choose... does it matter”	Multi-AZ Networking	Confusion choosing NAT subnet	Recommended match AZ or use one NAT for cost
33	“where is the add route option?”	Route Tables	User was on wrong page (association)	Guided to Route Tables → Routes → Edit → Add route
34	“where should i do this step... add NAT IP to MongoDB Atlas allowlist”	Atlas Security	Needed allowlisting	Guided in Atlas: Network Access → IP Access List
35	“where is the elastic ip?”	NAT Detail	Needed NAT public IP for allowlist	Identified “Primary public IPv4 address” as EIP
36	“what should i do here” (Compass screenshot)	Atlas vs Compass	User in Compass not Atlas	Redirected to Atlas web UI for allowlist
37	“is this okay?” (Atlas allowlist)	Verification	Confirmed allowlist	Confirmed correct; warned about 0.0.0.0/0 risk

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38	“what is the lambda code?”	Code Delivery	Needed exact Lambda code	Provided full Node.js export code + package.json deps
39	“where should i put this code?”	Packaging	Needed placement and zip process	Created lambda-export/ folder + build script + zip
40	“didn't u say node_modules should be in it?”	Deployment Artifact	Clarify node_modules requirement	Confirmed node_modules in zip but not committed to Git
41	“check all the files that should be in the zip are there?”	Validation	Ensure zip correctness	Verified zip contents include index.js/package.json/node_modules
42	“i have uploaded the zip file now what should i do?”	Deployment	Needed next steps post upload	Steps: env vars, S3 permissions, NAT routes, test
43	“should i deploy after uploading the zip?”	Lambda Behavior	Confusion about “deploy”	Clarified: upload + save config is enough
44	“from where on mongodb atlas can i find [MONGODB_DB]”	Atlas UI	Needed DB name	Guided to Data Explorer; found DB name test
45	“COLLECTIONS = orders,products,users,suppliers is this correct”	Export Config	Confirm correct collections	Confirmed from Data Explorer
46	“what is timeout and memory for?”	Lambda Ops	Understanding run time limits	Explained timeout/memory impact
47	“what should i do in this [Test Event]”	Lambda Testing	Needed to run test	Guided: create test event, run, check CloudWatch/S3

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48	“please [explain] the whole process... setup for Lambda function for migration of data”	Documentation	End-to-end process recap	Summarized full setup process
49	“now i want to connect aws glue with my data in S3... later using athena”	Data Analytics	Needed Glue/Athena integration	Designed S3→Glue→Athena workflow
50	“start with the first step with option B”	Data Lake Layout	Needed better folder layout	Planned per-collection folders
51	“update the index.js... rebuild the zip file for me”	Implementation	Needed code change and packaging	Updated key path to exports/<collection>/... and rebuilt zip
52	“guide me to step 2”	Process Guidance	Needed next step	Re-upload zip + run to generate folders
53	“done”	Progress	Confirmed Athena results location done	Moved to Glue setup
54	“for 4.2 is this correct?”	Glue Crawler	Validate S3 crawler path	Confirmed correct S3 path and settings
55	“where should i select the database...”	Glue Tables View	UI confusion	Explained database filter and crawler dependency
56	“summarize steps... connect S3 to glue and athena”	Documentation	Interim recap	Summarized completed steps
57	“if i add ALB would it cause issue with API gateway”	Architecture	Service overlap confusion	Explained ALB vs API Gateway patterns

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58	“architecture diagram not correct ... what changes”	Architecture	Correctness vs implementation	Recommended corrections and whether ALB needed
59	“backend on ec2... lambda export... help with flow”	Architecture	Finalize correct flows	Provided two-branch architecture: app flow + analytics flow

Chatgpt prompt: <https://chatgpt.com/share/695560c2-0d60-8007-88ae-12a8bab4fc91>

Gemini prompts:

- <https://gemini.google.com/share/b53d11274b4b>
- <https://gemini.google.com/share/a6ab8cb21768>
- <https://gemini.google.com/share/340edd35c262>

Topic Index (Grouped by Category)

prompts grouped by theme.

A) Application Features & Bug Fixing

- Restock integration (DB update)
- Orders create/delete issues
- Admin access controls
- Theme (light/dark)

B) Authentication & Deployment Debugging

- Amplify login issue, base URL /api fix
- CORS considerations
- Production API URL configuration

C) AWS Deployment & Infrastructure

- S3 creation settings
- Lambda runtime + packaging + env vars
- VPC + private subnets + NAT gateway

- Atlas IP allowlisting
- CloudWatch logs/testing workflow

D) Analytics Pipeline

- S3 export folder structure (Option B)
- Glue crawlers + Data Catalog
- Athena query results bucket